COM111 Tutorial 5a

1. For the following pairs of signals, determine if v_1 leads or lags v_2 and by how much?

   (a) $v_1 = 10\cos(5t - 20°)$, $v_2 = 8\sin 5t$

   (b) $v_1 = 19\cos(2t + 90°)$, $v_2 = 6\sin 2t$

   (c) $v_1 = -4\cos 10t$, $v_2 = 15\sin 10t$

2. What is the instantaneous voltage across a 2-μF capacitor when the current through it is $i = 4\sin(10^6 t + 25°)$ A?

3. What value of ω will cause the v_o, in Fig. 1 to be zero?

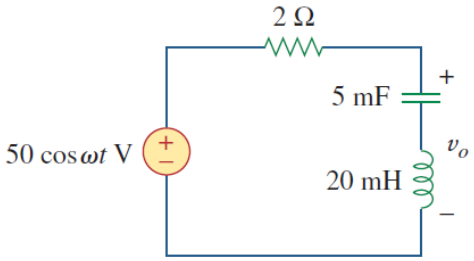


Fig. 1

4. For the circuit shown in Fig. 2, find Z_{eq} and use that to find current I. Let ω = 10 rad/s.

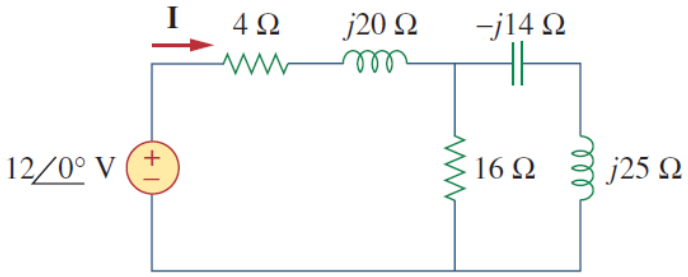


Fig. 2

5. In the circuit of Fig. 3, determine the value of $i_s(t)$.

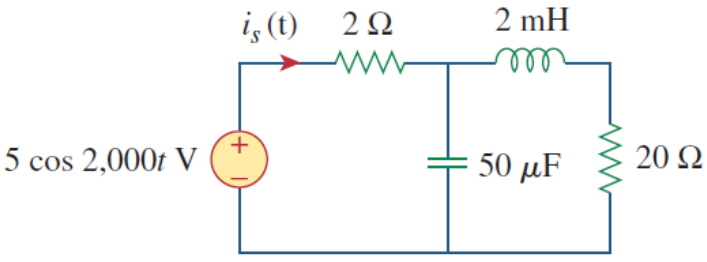


Fig. 3

6. For the network in Fig. 4, find Z_{in}. Let $\omega = 10$ rad/s.

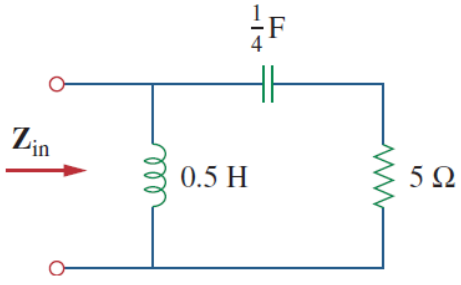


Fig. 4

7. Determine $\mathbf{Z}_T$ and $\mathbf{I}$ for the circuit in Fig. 5.

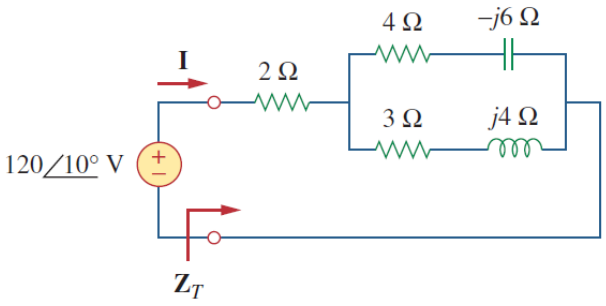


Fig. 5

8. Refer to the RC circuit in Fig. 6.
   (a) Calculate the phase shift at 2 MHz.
   (b) Find the frequency where the phase shift is 45°.

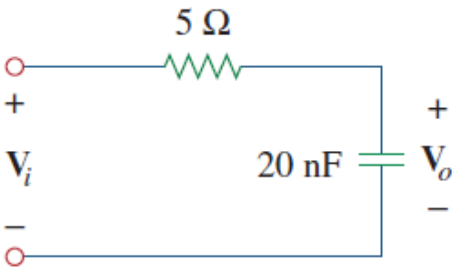


Fig. 6